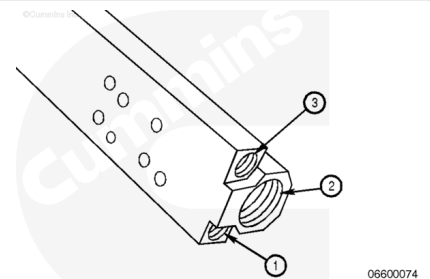


## Trainings ▾

### General Information with Mechanically Actuated Injector

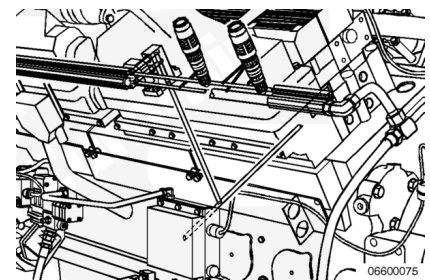
The QSK45 and QSK60 use a QSK fuel system that requires a fuel manifold on each bank with three fuel circuits. The three circuits are:

1. Rail
2. Drain
3. Timing.



#### **⚠ WARNING ⚠**

**Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to avoid severe personal injury or death when working on the fuel system.**

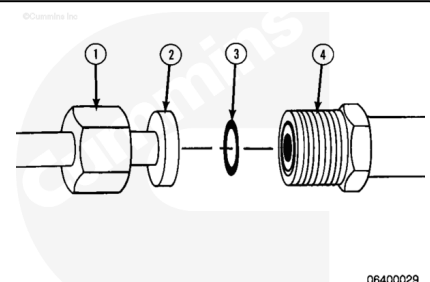


Fuel is delivered to the fuel blocks at the center of the engine, through drillings inside of the block, from the electronic control valve assembly.

The QSK45 and QSK60 use face seal o-ring fittings, except for the fuel filter inlet and the fuel drain connection that are 37 degree flare fittings.

Listed below are the components used in face seal o-ring fittings.

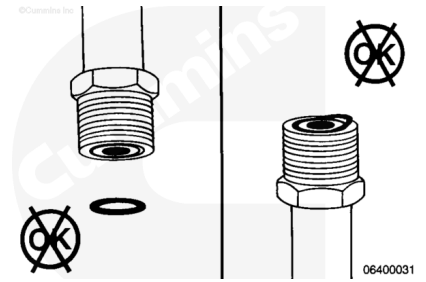
1. Tube nut
2. Tube flare flat
3. O-ring
4. Fitting.



#### **⚠ CAUTION ⚠**

**If Lubriplate® is not used, the o-ring can fall out or get pinched during assembly and cause a fuel leak.**

Hold the o-ring in position during assembly using Lubriplate® 105, or equivalent.

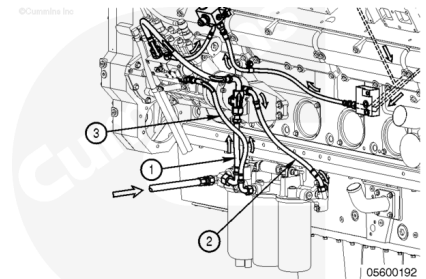


## with Electronically Actuated Injector

The fuel supply lines are those that transfer fuel from the fuel filter head to the gerotor pump, and the fuel pump. These are flexible hoses with either banjo fittings or o-ring face seals to prevent leakage.

The fuel flow for the fuel supply lines consists of the following:

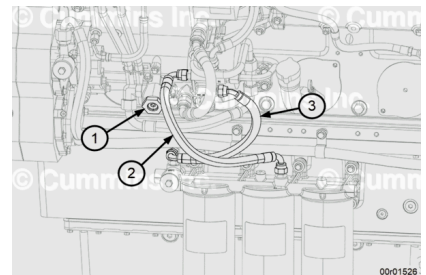
1. From first stage filter to gerotor pump
2. Gerotor output to second stage filters
3. From second stage filters to fuel pump.



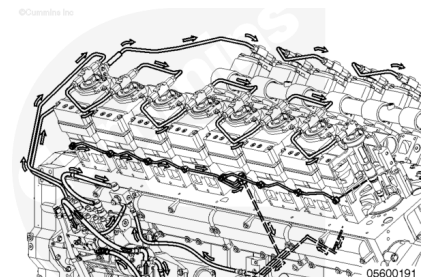
The Low NOx emission QSK60 uses a 2200 bar pump applied at 1600 bar. The fuel supply lines are those that transfer fuel from the fuel filter head to the gerotor pump, and the fuel pump. These are flexible hoses with either banjo fittings or o-ring face seals to prevent leakage.

The fuel flow for the fuel supply lines consists of the following:

1. From first stage filter to gerotor pump
2. Gerotor output to second stage filters
3. From second stage filters to fuel pump.

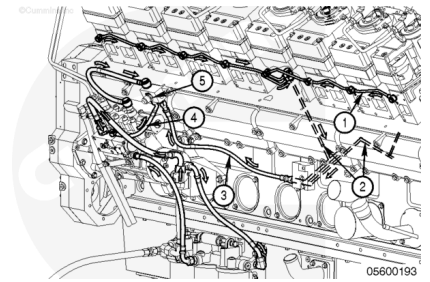


The injector supply lines carry the pressurized fuel from the fuel pump to the injectors. The injector supply lines are double wall pipe. The inner pipe carries the pressurized fuel to the injectors. The outer wall pipe is a protective shield so if there is a fuel leak it will direct the fuel to the vent line located on the side of the fuel pump.



The fuel return lines located on the cylinder head assemblies take fuel that was **not** injected from the injector and is returned to the vehicle fuel tank through the injector return manifold and the fuel manifold. The fuel return line on the mechanical dump valve takes excess fuel from the fuel pump and returns

it to the vehicle fuel tank. The inlet metering valve has a line which take excess fuel and return it back to the inlet of the gerotor pump. The recirculation provides constant pressure at the Fuel Pump Actuator to prevent fuel starvation to the pump. A second line running from the fuel pump actuator to the fuel return manifold is an air bleed to purge air from the fuel system (after fuel filter change).



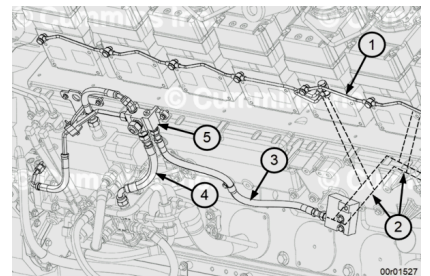
The fuel flow for the fuel return lines consists of the following:

1. Fuel return line from cylinder heads
2. Fuel return cavity through engine block to injector return manifold (mounted on the left side middle of the block)
3. Fuel return line from injector return manifold
4. Fuel return line from fuel pump actuator
5. Fuel manifold.

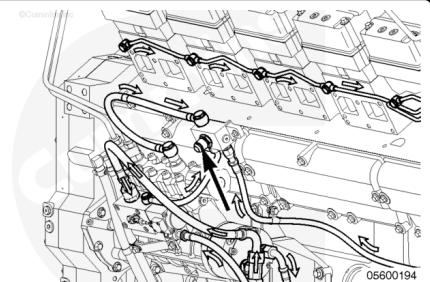
The Low NOx emissions QSK has a different routing for fuel drain lines.

The fuel flow for the fuel return lines consists of the following:

1. Fuel return line from cylinder heads
2. Fuel return cavity through engine block to injector return manifold (mounted on the left side middle of the block)
3. Fuel return line from injector return manifold
4. Fuel return line from fuel pump actuator
5. Fuel manifold.

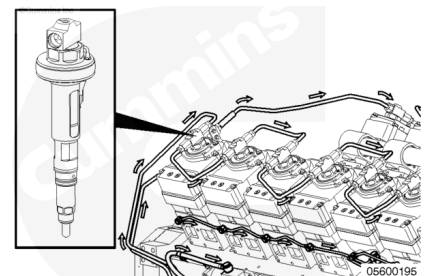


The fuel manifold provides one point where all the fuel returns connect and the fuel is returned to the vehicle fuel tank. The fuel manifold has an air bleed fitting for the fuel return line from the fuel pump actuator. This air bleed fitting is a check valve that opens at 5 bar [72 psi]. This limits flow rates out of the system at higher pressures and minimizes any flow loss during cranking and startup.



The injectors are made of three major components:

- A tee-connector which provides threaded ports for the injector supply line connection.
- An edge filter to collect debris, an orifice, and a flow limiter to help maintain a constant supply pressure in the injector.
- The injector body provides space to be an accumulator for the pressurized fuel. The electrical lead with connector, is



plugged into the main engine harness. The electronically controlled solenoid is used to operate the needle. The ECM commands the solenoid to control fuel injection.

A fuel cooler in the return fuel circuit is available as an option. The fuel is cooled using low temperature aftercooling (LTA) coolant.

